

```
nlr2007@NLR:~$ cat practical1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(){

char cmd[100];

printf("Enter Linux command: ");
scanf("%s", cmd);
pid_t pid = fork();
if(pid==0){
printf("Child PID : %d\n",getpid());
execlp(cmd, cmd, NULL);
perror("Execution Failed");
}

else if(pid>0){
printf("Parent PID : %d\n", getpid());
wait(NULL);
printf("child process completed.\n");
}
else {
printf("Fork Failed\n");
}
return 0;
}
```

Outputs:

```
nlr2007@NLR:~$ ./p1
Enter Linux command: uname
Parent PID : 723
Child PID : 726
Linux
child process completed.
nlr2007@NLR:~$ |
```

```

nlr2007@NLR:~$ ./p1
Enter Linux command: lscpu
Parent PID : 730
Child PID : 731
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1334U
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               4992.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr p

```

child process completed.

```

nlr2007@NLR:~$ ./p1
Enter Linux command: lsblk
Parent PID : 745
Child PID : 746
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 388.6M  1 disk
sdb  8:16   0   186M  1 disk
sdc  8:32   0    2G    0 disk [SWAP]
sdd  8:48   0    1T    0 disk /mnt/wslg/distro
/

```

child process completed.

```
nlr2007@NLR:~$ |
```

```

nlr2007@NLR:~$ ./p1
Enter Linux command: ps
Parent PID : 752
Child PID : 753
PID TTY          TIME CMD
410 pts/0        00:00:00 bash
752 pts/0        00:00:00 p1
753 pts/0        00:00:00 ps

```

child process completed.

```
nlr2007@NLR:~$ |
```

2. copying the contents from one file to another file

```
cannot open source file
1 nlr2007@NLR:~$ cat practical2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main(){
int src, dest;
char buffer[1024];
int bytes;
src = open("input.txt", O_RDONLY);
if(src<0){
printf("Cannot open source file\n");
return 1;
}

dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
write(dest, buffer, bytes);
}

close(src);
close(dest);

printf("File copied successfully. \n");

return 0;
}
nlr2007@NLR:~$ |
```